An Exact Real Quantum Latin Square of Order Six and Cardinality 29

Unsolved Labs Research Release R011

Abstract

We give an explicit real quantum Latin square of order six with cardinality 29. The construction is defined over a quadratic number field Q(tau), where tau is the positive root of 123201 tau^2 - 202800 tau - 93848 = 0. Thirty off-diagonal directions are constructed in R^5; after normalization they form orthonormal bases in every punctured row and column, and a common orthogonal diagonal vector extends the array to a quantum Latin square in R^6. Exact projective comparison shows that the thirty off-diagonal cells determine twenty-eight rays, so the common diagonal ray gives total cardinality 29.

The accompanying repository contains a standard-library exact reconstruction, a frozen exact coordinate certificate checked by a smaller data-driven verifier, and a separately written SymPy reconstruction. A distinct public cardinality-29 construction by Das and Kumar appeared on arXiv before the R011 repository was initialized; accordingly, no priority claim is made here.

1. Introduction

A quantum Latin square (QLS) is a vector-valued analogue of an ordinary Latin square: its entries are unit vectors and every row and column is an orthonormal basis. The notion was introduced by Musto and Vicary and is connected to unitary error bases and quantum combinatorial design.

For order six, recent work has clarified the attainable cardinalities, where cardinality means the number of entry rays after global phase is identified. Xu's 4 August 2026 revision left cardinality 29 as the remaining unresolved value in the spectrum described there. Das and Kumar posted an explicit real cardinality-29 construction on 12 August 2026. The R011 repository was initialized on 14 August 2026, so this artifact does not claim priority for resolving that gap.

Main theorem. There exists a real quantum Latin square of order 6 with cardinality 29.

2. Definitions and punctured-array reduction

An order-n QLS is an n by n array of unit vectors in C^n whose rows and columns are orthonormal bases. Its cardinality is the number of rays represented by its entries, where nonzero scalar multiples are identified.

We construct a punctured 6 by 6 array of nonzero directions in R^5. Normalize every off-diagonal direction v_ij and embed it as (v_ij,0) in R^6. Put the common vector f=(0,0,0,0,0,1)^T on every diagonal cell. If every punctured row and column is an orthogonal basis of R^5, then adjoining f produces a real QLS(6). If the off-diagonal directions represent r rays, the full square has cardinality r+1.

3. Quadratic parameter and building blocks

Let e_0,...,e_4 be the standard orthonormal basis of R^5. Let tau be the positive root of

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P(T) = 123201 T^2 - 202800 T - 93848.
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Define the rational rotations

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a=(4 e_3+3 e_4)/5, p=(-3 e_3+4 e_4)/5, d=(5 e_1-12 e_2)/13, t=(12 e_1+5 e_2)/13, q=(8 t+15
e_4)/17, r=(15 t-8 e_4)/17.
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Split q=X+H and introduce orthogonal companions Y,U by

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X=(96/221)e_1+(12/17)p, H=(40/221)e_2+(9/17)a, Y=(12/17)e_1-(96/221)p,
U=(9/17)e_2-(40/221)a.
```

Their squared norms are sigma=33552/48841 and rho=15289/48841, with sigma+rho=1. Put

```
S=rho X-sigma H; W=32 e_0-51 q, Z=51 e_0+32 q; B=51 rho e_0+32 H, C=32 e_0-51 H.
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Next define T_1=4e_0-3t and T_2=d+tau e_3. The vectors e_4,T_1,T_2 are mutually orthogonal. Let G be the orthogonal projection of X to their common orthogonal complement. Define

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K=(600 tau+1521)e_0 +1872e_1 +(2080 tau+780)e_2 +1920e_3.
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Exact reduction modulo P(tau)=0 gives K orthogonal to e_4,T_1,T_2,X, hence also K orthogonal to G. If g_U= and g_Y=, put L=g_U Y-g_Y U and M=g_Y rho Y+g_U sigma U. Let cr(v1,v2,v3,v4) be the five-dimensional generalized cross product and set N_3=cr(L,B,W,G), N_4=cr(M,C,X,K).

4. The punctured array

The off-diagonal directions are arranged as follows; a dash denotes a diagonal cell reserved for the common vector f.

```
V = [ - e0 e1 e2 e3 e4 ] [ 3e0+4t - -p a d T1 ] [ tau d-e3 q - e0 r T2 ] [ N3 L B - W G ] [
N4 M C X - K ] [ W S U Y Z - ]
```

All thirty directions are nonzero in Q(tau)^5. The repository freezes their exact reduced coordinates a+b tau in a deterministic JSON certificate.

5. Orthogonality

The rational rotations immediately give the first three rows. The decomposition q=X+H, together with the orthogonal companion pairs (X,Y) and (H,U), supplies the relations required for row 5 and columns 1 through 4.

Column 5 consists of e_4,T_1,T_2,G,K. By construction G is orthogonal to e_4,T_1,T_2, while K is orthogonal to e_4,T_1,T_2,X. Since G is the projection of X into the common orthogonal complement, K is orthogonal to G.

The remaining scalar closures are exactly where the quadratic parameter enters. They reduce to multiples of P(tau):

```
= 4608 P(tau) / [206353225 (1+tau^2)] = 0, = -154607616 (200 tau+507) P(tau) /
[171334463657825 (1+tau^2)] = 0.
```

The definitions of L,M,N_3,N_4 force the remaining row-3 and row-4 products to vanish. Column 0 can also be recovered by the frame identity: the normalized punctured rows contribute 6 I_5, while columns 1 through 5 contribute 5 I_5, leaving frame operator I_5 for column 0. The exact machine verifiers do not rely on this shortcut; they evaluate all 120 row/column pairwise inner products directly.

6. Exact ray count

For nonzero x,y in Q(tau)^5, projective equality is equivalent to vanishing of every 2 by 2 minor x_i y_j - x_j y_i. The release evaluates this criterion for all C(30,2)=435 off-diagonal pairs.

Exactly two off-diagonal ray classes repeat: V_01 is projectively equal to V_23, and V_34 is projectively equal to V_50. The first repetition is literally e_0=e_0 and the second is literally W=W. No other pair is proportional. Thus the thirty off-diagonal cells determine twenty-eight rays. The common diagonal vector is outside the embedded R^5, so it gives one additional ray. Therefore the total cardinality is 28+1=29.

7. Exact verification and trust boundary

The repository supplies three complementary exact checks. First, verify_qls6_card29.py reconstructs all directions from the displayed formulas using standard-library Fraction arithmetic in Q(tau). Second, certificate/qls6_card29_exact.json freezes the reduced exact coordinates, and verify_certificate.py checks the complete finite theorem without importing the construction code. Third, verify_qls6_card29_sympy.py independently implements the displayed formulas using SymPy polynomial reduction.

No decimal approximation is used as evidence for the theorem. The decimal vector export is for inspection only. R011 does not claim Lean verification; a faithful proof-assistant development would need to connect the quadratic field, all thirty exact vectors, all orthogonality and projective tests, and the final QLS/cardinality reduction. That development has not been completed.

8. Related work and scope

Musto and Vicary introduced quantum Latin squares in 2016. Recent 2026 work includes Xu's order-six constructions and broader cardinality-spectrum results. Das and Kumar publicly posted another real cardinality-29 QLS before the R011 repository was initialized. Their public construction is presented through a different displayed parameterization based on rational unit-circle parameters and orthogonal changes of basis.

The theorem here is intentionally narrow: one explicit real QLS(6) has cardinality 29. No uniqueness, classification, coordinate-optimality, or priority result is asserted.

Research provenance

This manuscript and repository form an Unsolved Labs public research artifact produced in a frontier-AI research workflow. No conventional human-authorship claim is implied by the organizational author line. Correctness is supported by the explicit mathematical argument and exact machine-verification artifacts, not by the provenance of the discovery process.

References

Musto and Vicary (2016). Quantum Latin squares and unitary error bases. Quantum Information and Computation 16, 1318-1332. DOI 10.26421/QIC16.15-16-4; arXiv:1504.02715.

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